

OSINT

The practice of collecting and analyzing information from publicly available sources to gain insights, knowledge, and understanding about a particular topic, individual, organization, or event.



OSINT Use Case Descriptions





CYBERSECURITY & THREAT INTELLIGENCE

- **Attack Surface Mapping:** Is used by companies with OSINT to see what an attacker sees to identify server leaks, exposing credentials of employees, or software that has been outdated and need of exposing.
- **Fraud Protection:** Dark web monitoring for stolen information or company log-in data being sold in hacker threads.
- **Vulnerability Assessment:** Tools such as Shodan are used to find publicly exposed IoT devices such as security cameras that lack password protection.
- **Phishing Defense:** Fake URLs and domains alike are tracked so they do not steal company data.



LAW ENFORCEMENT & PUBLIC SAFETY

- **Fugitive Recovery:** Locating suspects through their digital footprint, finding those who have skipped bail or committed crimes who are already in the database.
- **Anti-Trafficking:** Identifying and checking illicit networks by monitoring communications on encrypted sites.
- **Emergency Response:** Monitoring social media and other sites to identify hotspots for those in need in natural disasters and other emergencies.
- **Cold Case Closing:** Saved data, fingerprints, etc. To find evidence that was deleted or documented accounts of old witnesses.



JOURNALISM & HUMAN RIGHTS

- **Digital Forensics:** Using the metadata of leaked documents to verify authenticity before being published.
- **Disinformation Tracking:** Locating the origin of deepfakes or bots coordinated to alter data such as elections for example.
- **Verification of Conflicts:** Videos that have been geotagged are used by satellite to verify reports of military operations or human rights violations.
- **Environmental Monitoring:** Illegal logging of operations in areas that verified via satellite to verify authenticity.



MARKET EXPANSION & CORPORATE DUE DILIGENCE

- **Competitor Intelligence:** Analysis of competitor job postings, next moves, technology advancements, etc.
- **Brand Protection:** Identify fake domains and flag accounts pretending to be the company to avoid scam situations.
- **Market Trends:** Global analysis on which regions are ready for new products and set plans to launch them.
- **Vendor Vetting:** Analyzing public records to research partner reputations and trust history before signing business deals.

HOW OSINT IS USED



ADVANCED ONLINE SEARCHING (GOOGLE DORKING)

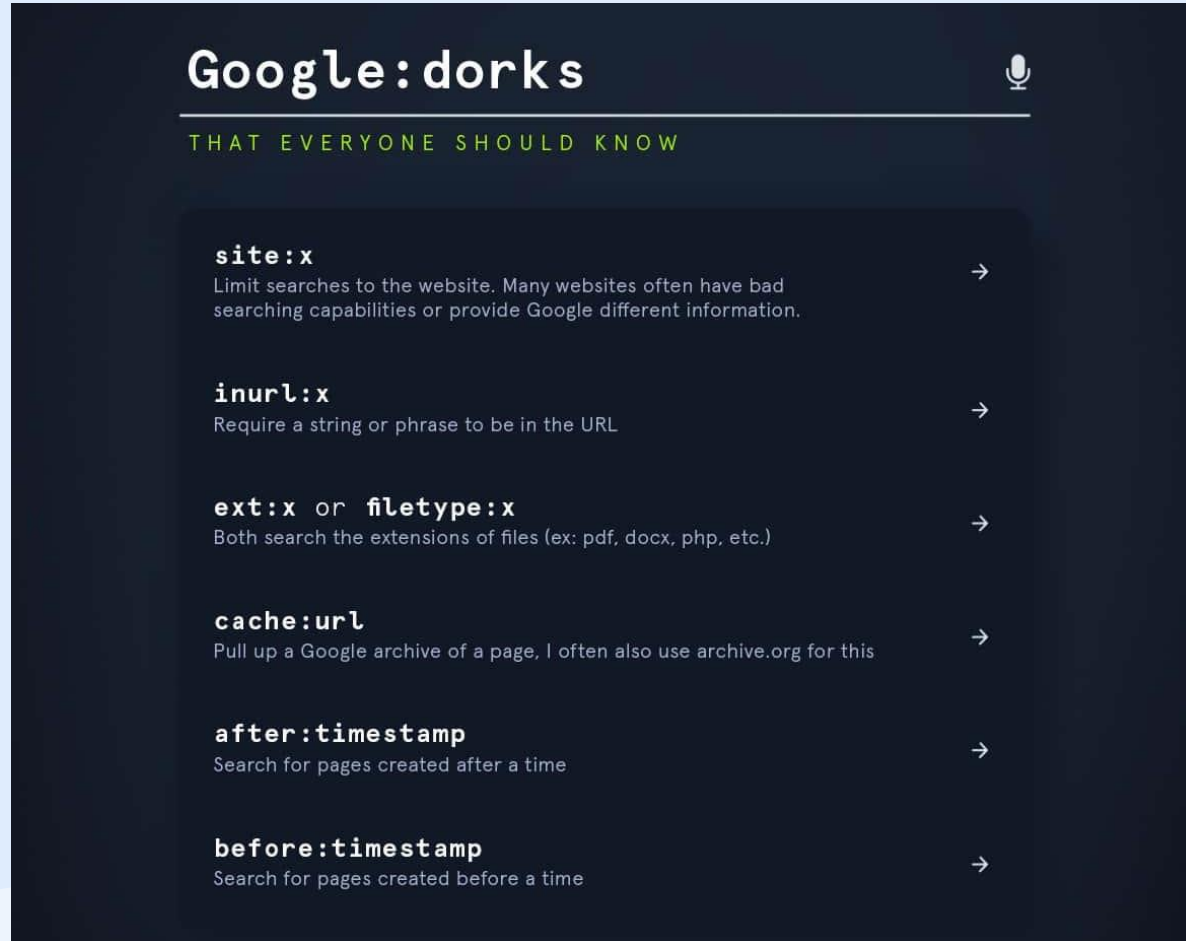
Key Functions:

site: Limits results to a specific domain

filetype: Searches for specific file extensions.

intitle: Uses keywords to find pages by title or tab.

inurl: Identifies keywords in URL strings.



ADVANCED ONLINE SEARCHING (GOOGLE DORKING) CONTINUED

Common Use Case:

Finding leaked password lists, exposed company directories, or reports that are not linked to the home page of corporations or government entities. Google Dorking is a surgical tool. Investigators can pull up any PDF file ever published to a specific site.





MARKETING STRATEGIES

- Implement personalized engagement strategies
- Tailor campaigns to resonate with specific demographics, fostering a sense of relevance and connection
- Collaborate with influencers and thought leaders to amplify our brand message

SOCMINT OR SOCIAL MEDIA INTELLIGENCE

The collection of information through social media sites

"Can be used to collect hidden or indirect but valuable information" (Chuahan, S., Panda, N., 2015).

Basic queries can produce basic info such as location photos and interests

More complex queries can uncover a person's likes, comments, captions, or an employer's company data

Social Media Search

Top Social Networks Search Results Dashboard

POTENTIAL ETHICAL CONSIDERATIONS



A word cloud on a dark blue background featuring various ethical and philosophical terms. The words are arranged in a dense, overlapping manner, with some words appearing larger and more prominent than others. The color palette for the words includes white, light blue, and medium blue. The background of the slide features a repeating pattern of stylized, overlapping leaf or petal shapes in shades of blue.

IDEALS CODE CONVENTIONALITIES CULTURE
RECTITUDE BELIEF AUTHENTICITY
TEMPERANCE LAW WORTHINESS STANDARDS
MOTIVATION RESPONSIBILITY TREATMENT SENSE
THEORY RESPECTABILITY
ETHICS **MORAL**
VALUES
TRUTH
HONOR
ETIQUETTE
FAITH
CITERIA
FAIRNESS
DECENCY
INNOCENCE
ACCEPTANCE
SOCIAL
EQUALITY
FAIRNESS
CHARACTERS
PERFORMANCE
RESPECT
PHILOSOPHY
WORTHINESS
PURPOSE
INTEGRITY
RULES
ATTITUDE
STANDARDS
CHARACTER
HYPOCRISY
VIRTUE
EQUALITY
DECENCY

POTENTIAL ETHICAL CONSIDERATIONS

Invasion of Privacy - "collecting information from various sources, including social media, online databases, and public records can lead to exposure of personal information that was not intended to be public" (Bolen, S, 2024).

Consent & Context - "Information shared in one context can be taken out of another, leading to misinterpretation and potential harm. The notion of informed consent is often absent in broad OSINT collection" (Mianji, S, 2025).

Potential for Misuse - "In the wrong hands, OSINT techniques can be used for malicious purposes, including doxing, stalking, and targeted harassment" (Mianji, S, 2025).

Accuracy & Misinformation - "Social media is a breeding ground for misinformation and disinformation. OSINT practitioners must be adept at verifying sources and corroborating information to avoid acting on false intelligence. The sheer volume of data – the "signal vs. noise" problem – further complicates this" (Mianji, S, 2025).

WHAT IS DOMAIN RESEARCH IN OSINT?

Domain research is the systematic investigation into the particular field or sector in which a business, project, or website operates. It helps identify the landscape, key players, target audience, competitors, and industry trends.

Purpose:

To understand the niche market or industry standards

To identify opportunities and threats

To analyze competitors' strengths and weaknesses

Benefits:

Better positioning and differentiation

More targeted marketing efforts

Reduced risks by understanding the competitive landscape

KEY OBJECTIVES OF DOMAIN RESEARCH



IDENTIFY DOMAIN
OWNERSHIP AND
REGISTRATION DETAILS



DISCOVER HOSTING
AND INFRASTRUCTURE
DETAILS



DETECT MALICIOUS OR
SUSPICIOUS ACTIVITY



UNDERSTAND DOMAIN
RELATIONSHIPS AND
HISTORY

Tools for Domain Research

WHOIS LOOKUP (E.G., ICANN, DOMAINTOOLS)

DNS RECORDS (E.G., NSLOOKUP, DIG)

REVERSE IP LOOKUP

SSL/TLS CERTIFICATE ANALYSIS

PASSIVE DNS DATA

WEB ARCHIVE (WAYBACK MACHINE)

STEP-BY-STEP DOMAIN RESEARCH PROCESS

Initial domain identification

WHOIS data collection and analysis

DNS record investigation

Hosting infrastructure analysis

SSL certificate examination

Historical data review

Cross-referencing with other sources

CHALLENGES/BEST PRACTICES

Common Challenges in Domain Research

- Privacy protection services
- Dynamic IP addresses
- Domain cloaking
- Data accuracy and reliability

Best Practices

- Use multiple tools for cross-verification
- Regularly monitor domain changes
- Maintain updated knowledge of TLD and privacy services
- Respect legal boundaries and privacy laws

THE IMPORTANCE OF OSINT IN DOMAIN RESEARCH



Domain research in OSINT is crucial for understanding a target through publicly available info. It helps uncover key assets, relationships, and vulnerabilities, providing context and supporting strategic decisions. Overall, it streamlines investigations and enhances the accuracy of insights.



Effective domain research enhances threat detection and investigation by revealing malicious assets, connections, and vulnerabilities. It helps identify suspicious domains, track attacker infrastructure, and uncover hidden links, enabling quicker response and better understanding of potential threats.



Practicing regularly and staying updated on tools and techniques is essential for effective OSINT and threat detection. Continuous learning enhances skills, keeps you ahead of evolving threats, and ensures you can leverage the latest resources for more accurate and efficient investigations. Keep exploring and updating your knowledge to stay effective and confident in your work.

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Thank you

ALEXIS GANT

GAVIN BENNETT

BRITTANY KITTLE

IST 110 GROUP 6

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